

# A Dual-Model Machine Learning Framework for PM<sub>2.5</sub> Prediction and Air Quality Classification

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**Abstract**—Air pollution is one of the major issues in India. Tiny particulate matter (PM<sub>2.5</sub>) consists of airborne particles smaller than 2.5 micrometers which can even penetrate the human respiratory system, thus are most harmful components of air pollution. This research develops predictive models using advanced ensemble learning methods-XGBoost, LightGBM, and CatBoost-along with a voting ensemble, leveraging multipollutant and temporal data from 10 major Indian cities, collected from the Central pollution control board(CPCB) for 2024. The dataset incorporates air pollutants, combined with daily and hourly temporal features. After preprocessing, rigorous cleaning and feature engineering, the models were trained for both regression (PM<sub>2.5</sub> concentration prediction) and classification (Air Quality Index category). Among regressors, LightGBM achieved superior performance with an  $R^2$  of 0.8649, RMSE of 21.97 and an MAE of 8.63. The voting ensemble classifier obtained the highest classification accuracy of 83.16% with a macro F1-score of 0.77, performing well in the “Good” and “Satisfactory” categories but showing challenges in extreme classes due to imbalance. These results demonstrate an interpretable framework that provides accurate PM<sub>2.5</sub> estimates and reliable AQI classifications, supporting public health awareness and enabling timely policy interventions.

**Index Terms**—PM<sub>2.5</sub> Prediction, Air Pollution, Ensemble Learning, Air Quality Classification, Temporal Features

## I. INTRODUCTION

Air pollution is a critical environment challenge worldwide, with fine particulate matter (PM<sub>2.5</sub>) recognized as one of the most harmful pollutants. Due to its small diameter ( $\leq 2.5 \mu m$ ), PM<sub>2.5</sub> can reach lung tissues, enter the bloodstream, triggering severe respiratory and cardiovascular complications. According to the “World Health Organization” (WHO) [1], recommended limits for PM<sub>2.5</sub> are an annual average not higher than  $5 \mu g/m^3$ , and daily average should stay below  $15 \mu g/m^3$ .

Exposure to PM<sub>2.5</sub> concentrations above safe limits causes many premature deaths in India, accounting for nearly quarter of total mortality [2]. Rapid urbanization, vehicular emissions, and industrial growth have drastically increased ambient PM<sub>2.5</sub> levels, with over twenty major cities frequently exceeding safe

thresholds [3]. These alarming levels highlight the urgent need for predictive models to monitor, forecast, and mitigate PM<sub>2.5</sub> exposure through timely interventions and informed policy actions.

Recent studies show that PM<sub>2.5</sub> pollution affects both urban and rural areas, disproving the notion that it is solely an “urban problem” [4]. This highlights the widespread health risks across India and the need for accurate PM<sub>2.5</sub> prediction and remark classification to support timely health advisories and policy action. Traditional models struggle to capture nonlinear interactions among emissions, weather and pollutants [5], limiting real-time prediction accuracy. They also discard incomplete or extreme data, reducing realism [4]. Hence, advanced ML models are needed to uncover hidden patterns.

Here this study presents a machine learning-based framework for urban air quality prediction using CPCB 2024 dataset of hourly and daily observations [6]. Missing values were replaced with daily averages, while genuine outliers were retained to capture pollution spikes. A multi-model pipeline combining gradient boosting methods was used for both regression and classification. LightGBM achieved the highest  $R^2=0.865$  for PM<sub>2.5</sub> prediction, and soft-voting ensemble of XGBoost, LightGBM, and CatBoost classifiers achieved 83.2% accuracy after SMOTE-Tomek resampling. The framework delivers precise PM<sub>2.5</sub> estimates and interpretable AQI insights by exclusively using ground-based CPCB data, pragmatic handling of missing values, and retaining outliers, thereby ensuring data integrity and supporting practical urban air quality forecasting and policy decisions in India.

## II. LITERATURE REVIEW

Rising PM<sub>2.5</sub> levels in India is posing health risk, making accurate prediction and AQI classification essential for its control and public health. Recent studies have applied ensemble boosting and imbalance-handling techniques. For instance, *Frontiers in Big Data* (2023) [7] introduced an ensemble

based AQI classification framework for chinese cities using meteorological indicators, comparing multiple models and class balancing method.

In [8], the author evaluated the performance of LightGBM, CatBoost, XGBoost, Random Forest, and other models to predict weekly  $PM_{2.5}$  peaks in Algiers, Algeria, with a particular focus on enhancing models interpretability. Likewise, Park et al. [9] proposed a spatiotemporal ensemble using multiple boost algorithms to estimate surface  $PM_{2.5}$  in South Korea. Research on urban air quality sensor network showed that removing outliers may overlook critical insights, thus limiting the robustness of the model [10].

Machine learning algorithms have used the SMOTE technique to balance minority classes and to improve performance [7], showing the importance of class balancing. In addition, SMOTE-Tomek combination further reduces imbalance and increases predictive accuracy in air quality classification [8], highlighting the effectiveness of resampling methods for robust and reliable results.

Gradient boosting models are widely known for strong performance in environmental prediction tasks. Thus are suitable for  $PM_{2.5}$  forecasting. The study [11] found gradient boosting model achieved high accuracy among ten regression algorithms, while [12] and [13] highlighted their robustness in modeling non-linear relationships, temporal dependencies, and complex feature interactions with SHAP-based interpretability. These studies support their use in  $PM_{2.5}$  prediction pipeline for both accuracy and explainability.

Integrating SMOTE with boosting models effectively reduces class imbalance [14]. While Hybrid SMOTE-XGBoost framework improves AQI prediction accuracy and robustness, while a Voting Ensemble via majority or probability averaging further reduces bias and enhance performance [15].

### III. METHODOLOGY

This research presents a full methodological framework adopted in study, at all key stages.

The hourly pollutants were aggregated into daily records for consistency. Preprocessing addressed missing values and encoded categorical features, while outliers were retained to reflect genuine pollution spikes. Class imbalance was handled using SMOTE-Tomek [16], ensuring balanced class representation.

Feature engineering included temporal breakdown of timestamps, application of cylindrical encoding, and generation of lag and rolling-average variables to capture both seasonal patterns and short-term variations in  $PM_{2.5}$ . The Ensemble methods (XGBoost, LightGBM and CatBoost) were tuned with hyperparameters and cross-validation to predict continuous  $PM_{2.5}$  values that were later mapped to AQI categories while soft-voting ensemble improved the robustness of classification.

Model performance was evaluated using regression metrics (RMSE, MAR,  $R^2$ ) and classification metrics (precision, recall, F1-score), supported by visual analyses of trends, feature importance, and errors offering deeper insights.

#### A. Data Collection

The raw dataset sourced from **CPCB( Central pollution control board)** [6], the national authority for responsible for air quality monitoring in India, had features like hourly measurement of key air pollutants with meteorological parameters which were collected for time range : **January 2024 to December 2024**, thus have full one year coverage. Selection of cities was based on diversity of geographical and environmental conditions across India for comprehensive air quality coverage [17]. The data were collected from **10 different monitoring Indian stations**, whose geographic distribution is depicted in figure 1, showing diversity in region. For data continuity, missing hourly values were imputed using corresponding daily averages.

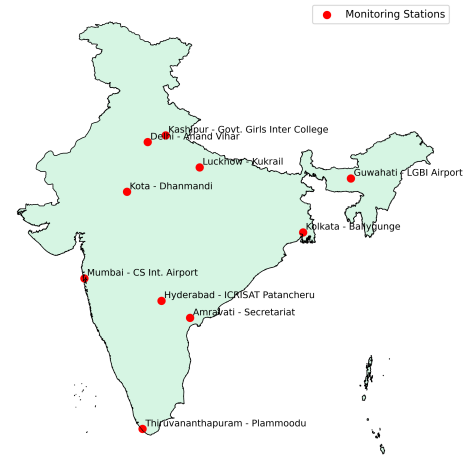


Fig. 1. Geographic locations: 10 monitoring stations of India used for study

#### B. Data Preprocessing

The dataset was obtained from CPCB [6] monitoring stations across ten Indian cities from different region contained **missing values and inconsistencies** that needed proper and systematic pre-processing prior to model development. Missing values in such dataset may arise due to some factors such as *temporary sensor malfunction at monitoring stations and manual recording errors*, both of these are common challenges in environmental data collection. Missing data were non-uniformly distributed across stations, also affected features varied significantly. For data quality, the study applied combination of statistical imputation, interpolation techniques, and feature specific handling strategies depending in characteristics of the station.

##### Data Cleaning :

In the rows where  $PM_{2.5}$  was absent, such rows are removed. Missing values were identified and addressed using imputation strategies. Different methods are used to handle missing, i.e, skewness correction followed by row-wise removal and consistency checks to ensure a clean and reliable dataset.

Missing value summary for each station with method used to handle it(see Table I).

TABLE I  
MONITORING STATIONS AND THEIR RESPECTIVE METHODS TO HANDLE MISSING VALUES.

| Station ID                                | Feature with Missing                                    | Method Used                                                            |
|-------------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------|
| Anand Vihar                               | WS ,NO <sub>2</sub> ,SO <sub>2</sub> ,CO,O <sub>3</sub> | Mean Imputation                                                        |
| Chhatrapati Shivaji International Airport | NH <sub>3</sub> , SO <sub>2</sub> ,WS,SR                | Log-transform for skewness+Median Imputation                           |
| Ballygunge                                | None                                                    | No handling required                                                   |
| Plammoodu                                 | None                                                    | No handling required                                                   |
| ICRISAT Patancheru                        | SO <sub>2</sub> ,                                       | Median (SO <sub>2</sub> ) & Linear Interpolation(Ozone)                |
| Dhanmandi                                 | BP                                                      | Interpolation                                                          |
| Govt. Girls Inter College                 | SO <sub>2</sub> , NH <sub>3</sub> , RH(%)               | Interpolation (linear fill after fillna)                               |
| Kukrail Picnic Spot-1                     | SR                                                      | Dropped rows with missing values in SR                                 |
| LGBI Airport                              | AT, CO, WS, BP, RH(%), WD, SR                           | Linear Interpolation (AT, CO, WS, BP) & Median Imputation (RH, WD, SR) |
| Secretariat                               | RH(%)                                                   | Median Imputation                                                      |

### C. Feature Engineering

Feature engineering was done to enhance model learning and capture temporal dependencies in air pollution data. Temporal features (year, month, day, day of week, and hour) were extracted from the timestamp. Cyclical temporal attributes were transformed into continuous representation using sine and cosine function which preserve periodic behaviour:

$$\text{month\_sin} = \sin\left(\frac{2\pi \times \text{month}}{12}\right), \quad \text{month\_cos} = \cos\left(\frac{2\pi \times \text{month}}{12}\right), \\ \text{hour\_sin} = \sin\left(\frac{2\pi \times \text{hour}}{24}\right), \quad \text{hour\_cos} = \cos\left(\frac{2\pi \times \text{hour}}{24}\right)$$

Such trigonometric encoding is commonly used in representation of periodic temporal features in time series learning. Lagged PM<sub>2.5</sub> values (1hour, 3hour) and rolling averages (3hour, 24hour) are used to identify temporal patterns and trends. For data consistency missing valued rows were removed.

*Method for outlier detection and analysis:*

- **Skewness:** Gives numerical values to check asymmetrical distributions.
- **Heatmap Correlation** is giving plot for multi collinear variables and to identify redundant or highly dependent features.
- **Scanning distribution plots:** Histogram with kernal density estimations(KDE) and skewness values give visualization to observe whether data is normally distributed or skewed.
- **Box Plots** for outlier detection and visualize spread across features.

### D. Feature Selection and Target Preparation

The dataset contained 17 attributes, including temporal, pollutant, meteorological, and geospatial features. A summary of these features is presented in Table II.

**Geospatial Feature Integration:** The dataset contained observations from ten monitoring stations across different Indian cities, as shown in Figure 1 . The latitude and longitude of each station were added as features to provide spatial context. These geospatial features helped the models learn regional differences in air quality and improved their prediction performance.

TABLE II  
FEATURES CONSIDERED FOR REGRESSION AND CLASSIFICATION TASKS.

| Category        | Features                                                                                 | Type       |
|-----------------|------------------------------------------------------------------------------------------|------------|
| Target Variable | PM <sub>2.5</sub> ( $\mu\text{g}/\text{m}^3$ )                                           | Continuous |
| Air Pollutants  | NO,NO <sub>2</sub> ,NO <sub>x</sub> ,NH <sub>3</sub> ,SO <sub>2</sub> ,CO,O <sub>3</sub> | Numeric    |
| Meteorological  | AT (°C), RH (%),WS (m/s),WD(°),SR (W/m <sup>2</sup> ),BP(mmHg)                           | Numeric    |
| Geospatial      | Latitude, Longitude                                                                      | Numeric    |
| Temporal        | Timestamp                                                                                | DateTime   |

**Target Preparation:** For regression tasks, the target variable was the continuous PM<sub>2.5</sub> concentration. For classification tasks, AQI categories were obtained from CPCB standards [6], which map PM<sub>2.5</sub> values into six classes: "Good", "Satisfactory", "Moderate", "Poor", "Very Poor", and "Severe".

### E. Exploratory Data Analysis

- **Day-Level Analysis:** Figure 2 shows PM<sub>2.5</sub> levels for a selected day, capturing short-term pollution events and allowing closer inspection of daily variability.
- **Station-Wise Markers:** Represents PM<sub>2.5</sub> values using different markers for each station, improving the clarity of overlapping points and emphasizing spatial heterogeneity in pollutant levels.
- Figure 3 shows clear spatial differences in air quality, with some stations showing larger interquartile ranges and more outliers, indicating higher variability and occasional spikes in pollution at those locations.

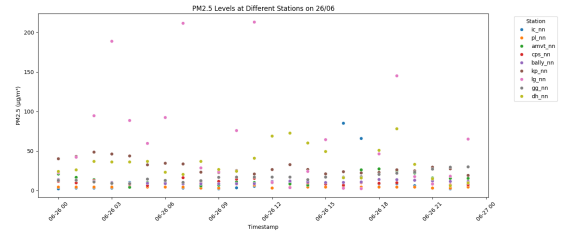


Fig. 2. PM<sub>2.5</sub> levels at different monitoring stations on selected date

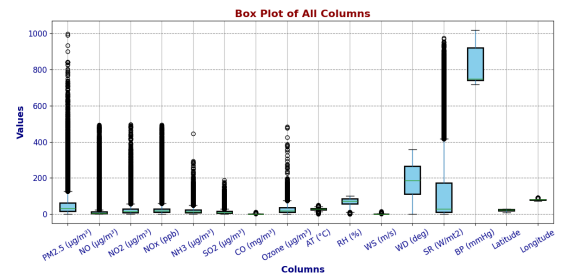


Fig. 3. Combined box plot of PM<sub>2.5</sub> across all monitoring stations

### F. Model Training

**1) Regression Models:** The primary objective of regression modeling was to predict continuous PM<sub>2.5</sub> concentrations. Three advanced ensemble learning models were employed: **XGBoost Regressor** [18], **LightGBM Regressor** [19] and

**CatBoost Regressor** [20]. All models were trained with tuned hyperparameters and a fixed random seed was applied to guarantee reproducibility.

- **XGBoost Regressor**

XGBoost Regressor [18], a fast and high performance gradient boosting framework. The model was trained with 300 estimators, a learning rate of 0.05, and maximum depth of 8. The histogram-based tree method was used to speed up on large datasets. The input features were standardized, and performance was evaluated using  $R^2$ , RMSE, and MAE.

- **LightGBM Regressor**

LightGBM Regressor [19], a gradient boosting method that grows trees leaf-wise, which can give higher accuracy and faster training on large datasets. The model was set up with 300 estimators, a learning rate of 0.05, and no limit on tree depth (-1), to allow full tree growth. The input features were standardized, and the same evaluation metrics were applied.

- **CatBoost Regressor**

CatBoost Regressor [20] uses ordered boosting and symmetric trees to handle categorical and numerical data effectively and lower the risk overfitting. The model was trained for 300 iterations with a learning rate of 0.05 and a tree depth of 8.

2) **Classification Models:** To provide clear air quality information, the continuous  $PM_{2.5}$  predictions were converted into six AQI categories: Good, Satisfactory, Moderate, Poor, Very Poor, and Severe. Models included XGBoost Classifier, LightGBM Classifier, CatBoost Classifier, and a Voting Ensemble combining all three.

- **Handling Class Imbalance**

The AQI categories were highly uneven, with “Good” and “Satisfactory” dominating the dataset. To fix this, SMOTE-Tomek [16] method was used on the training set to increase the rare classes and reduce the very common ones. A stratified train-split was used so that the test maintained the original distribution for unbiased evaluation.

- **XGBoost Classifier**

The XGBoost classifier was tuned using a randomized search across key hyperparameters, including the number of trees, learning rate, maximum depth, and subsampling ratios. XGBoost uses gradient boosting with second-order optimization and built-in regularization, which helps it learn complex non-linear patterns. The model was evaluated using accuracy, macro-averaged F1-score, and confusion matrices.

- **LightGBM Classifier**

LightGBM classifier training used a simpler hyperparameter search. The model relied on histogram-based decision tree learning, column-wise parallelization, and gradient-based one-sided sampling(GOSS), which together made training faster and more scalable while still keeping high accuracy on large datasets.

- **CatBoost Classifier**

CatBoost was chosen for its efficient handling of categorical features and robustness against overfitting. It was trained for

300 iterations with a learning rate of 0.05 and depth of 8, using ordered boosting to reduce overfitting and manage categorical interactions.

- **Voting Ensemble**

The final classification model was a Voting Ensemble combining XGBoost, LightGBM, and CatBoost with soft voting. This combined model achieved slightly higher **accuracy (83.15%)** and **macro F1-score (0.77)** than any single classifier, highlighting that combining different models improves performance. The Voting Ensemble ensured reliable AQI classification and worked as a strong setup for real-time air quality monitoring.

### G. Model Prediction

1) **Remark Classification:** Table IV shows the performance of regression models for continuous  $PM_{2.5}$  prediction. **LightGBM gave the best fit with  $R^2$  of 0.8649**, slightly better than **CatBoost (0.8588)** and **XGBoost (0.8556)**, meaning the models explain around 85–86% of the variance in  $PM_{2.5}$ . **RMSE and MAE for LightGBM were 21.97 and 8.63  $\mu g/m^3$** , respectively, providing interpretable error measures and highlighting the effectiveness of the feature engineering pipeline.

TABLE III  
IND-AQI CATEGORY AND  $PM_{2.5}$  BREAKPOINTS

| AQI Category | AQI Range | $PM_{2.5}$ Breakpoints ( $\mu g/m^3$ ) |
|--------------|-----------|----------------------------------------|
| Good         | 0–50      | 0–30                                   |
| Satisfactory | 51–100    | 31–60                                  |
| Moderate     | 101–200   | 61–90                                  |
| Poor         | 201–300   | 91–120                                 |
| Very Poor    | 301–400   | 121–250                                |
| Severe       | 401–500   | 251+                                   |

The National Air Quality Index (AQI) [6] for a pollutant P is derived using a piecewise linear function that converts continuous  $PM_{2.5}$  values into AQI categories for public health guidance, is as follows:

$$AQI_P = \frac{I_{HI} - I_{LO}}{BP_{HI} - BP_{LO}} \times (C_P - BP_{LO}) + I_{LO} \quad (1)$$

Where:

$AQI_P$  – AQI for pollutant P

$BP_{HI}$  – Breakpoint  $\geq C_P$

$BP_{LO}$  – Breakpoint  $\leq C_P$

$I_{HI}$  – AQI value corresponding to  $BP_{HI}$

$I_{LO}$  – AQI value corresponding to  $BP_{LO}$

$C_P$  – Concentration of pollutant P

2) **Model Evaluation Metrics:** The performance of regression and classification models was checked using standard metrics. For regression, we used  **$R^2$ , Root Mean Square Error (RMSE), and Mean Absolute Error (MAE)**. For classification, the metrics were **Accuracy, Macro F1-score, and Weighted F1-score**.

#### Regression Performance:

Table IV summarizes the performance of the three regression models used to predict continuous  $PM_{2.5}$  concentrations. LightGBM Regressor achieved the highest  $R^2$  of 0.8649, indicating strong predictive capability.

TABLE IV  
REGRESSION MODEL PERFORMANCE

| Model              | $R^2$  | RMSE ( $\mu\text{g}/\text{m}^3$ ) | MAE ( $\mu\text{g}/\text{m}^3$ ) |
|--------------------|--------|-----------------------------------|----------------------------------|
| XGBoost Regressor  | 0.8556 | 22.72                             | 8.39                             |
| LightGBM Regressor | 0.8649 | 21.97                             | 8.63                             |
| CatBoost Regressor | 0.8588 | 22.46                             | 8.87                             |

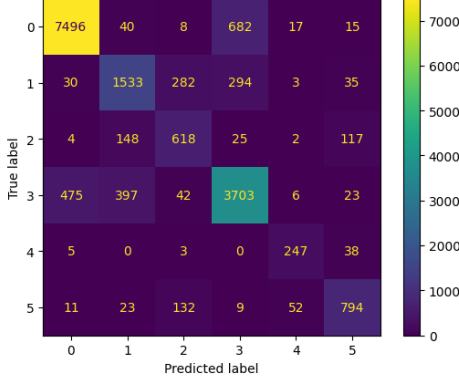


Fig. 4. Confusion matrix

#### Classification Performance:

Table V presents the performance of the classification models for predicting AQI category. The Voting Ensemble Classifier beat the individual models in both accuracy and F1-score, providing the benefit of combining different learners for handling imbalanced data (Table VI). Its classification report and confusion matrix (Figure 4) shows high precision and recall for “Good” and “Satisfactory” (**F1: 0.90, 0.79**), reliable performance for “Moderate” and “Poor” (**F1 : 0.71, 0.62**), and strong results for minority classes “Severe” and “Very Poor” (**F1: 0.80, 0.77**), demonstrating the effectiveness of SMOTE-Tomek.

TABLE V  
CLASSIFICATION MODEL PERFORMANCE

| Model                      | Accuracy | F1-scores                   |
|----------------------------|----------|-----------------------------|
| XGBoost Classifier         | 0.8261   | —                           |
| LightGBM Classifier        | 0.8315   | —                           |
| CatBoost Classifier        | 0.8212   | —                           |
| Voting Ensemble Classifier | 0.8315   | Macro: 0.77, Weighted: 0.83 |

TABLE VI  
VOTING ENSEMBLE CLASSIFIER REPORT

| AQI Category          | Precision | Recall | F1-score | Support |
|-----------------------|-----------|--------|----------|---------|
| Good (0–50)           | 0.94      | 0.91   | 0.92     | 8258    |
| Satisfactory (51–100) | 0.79      | 0.80   | 0.79     | 4646    |
| Moderate (101–200)    | 0.71      | 0.70   | 0.71     | 2177    |
| Poor (201–300)        | 0.57      | 0.68   | 0.62     | 914     |
| Very Poor (301–400)   | 0.78      | 0.77   | 0.77     | 1021    |
| Severe (401–500)      | 0.75      | 0.87   | 0.80     | 293     |
| Accuracy              | -         | -      | 0.83     | 17309   |
| macro average         | 0.75      | 0.79   | 0.77     | 17309   |
| weighted average      | 0.84      | 0.83   | 0.83     | 17309   |

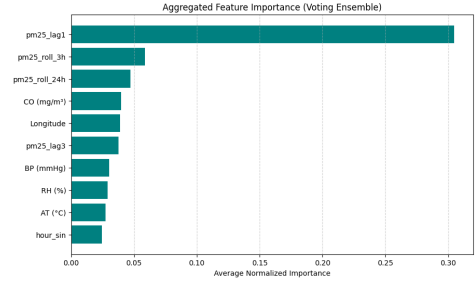


Fig. 5. Feature Importance

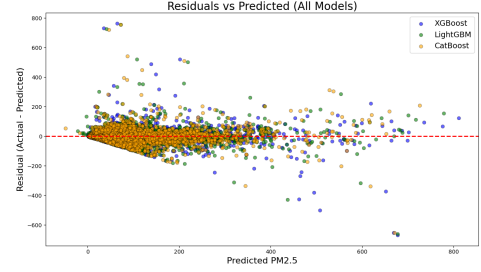


Fig. 6. Residual Analysis

#### H. Visualization

**Confusion Matrices:** Figure 4 shows strong values along the diagonal, confirming accurate classification across AQI categories and balanced ensemble performance.

**Feature Importance:** Figure 5, identifies recent  $\text{PM}_{2.5}$  values and short-term rolling averages as dominant predictors, with meteorological and temporal variables contributing marginally.

**Residual Analysis:** Residual plots (see figure 6) indicate minimal errors at low  $\text{PM}_{2.5}$  level, and greater dispersion during pollution peaks.

**AQI vs  $\text{PM}_{2.5}$ :** Figure 7 validates consistent translation of predicted  $\text{PM}_{2.5}$  values into CPCB-based AQI categories, ensuring standardized interpretation.

## IV. RESULTS

The dual-model framework for  $\text{PM}_{2.5}$  regression and AQI classification showed strong results: Table IV shows that LightGBM achieved highest  $R^2$  (0.865), followed by CatBoost(0.859) and XGBoost(0.856). Table V shows that the Voting Ensemble reached 83.2% accuracy, exceeding XGBoost(82.6%) and CatBoost(82.1%). High F1-scores for “Good” (0.92), “Satisfactory” (0.79), and balanced minority classes (“Severe”: 0.80, “Very Poor”: 0.77) confirm SMOTETomek’s success in handling imbalance, with minor confusion between “Moderate” and “Poor”(Table VI).

## V. CONCLUSION

This study introduced a dual-model framework for predicting  $\text{PM}_{2.5}$  concentrations and classifying Air Quality Index categories across major Indian cities using CPCB ground monitoring data. Ensemble boosting models—XGBoost, LightGBM, and CatBoost—were able to capture temporal patterns as well as pollution-driven variations in AQI, with



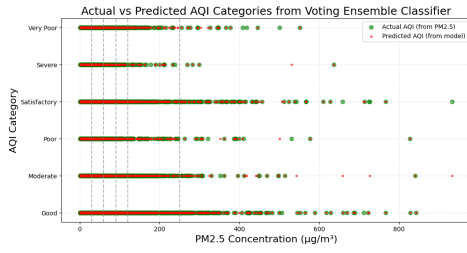


Fig. 7. AQI vs PM<sub>2.5</sub>

LightGBM providing the highest predictive accuracy among all the models tested. The Voting Classifier showed strong performance in correctly identifying “Good” and “Satisfactory” AQI categories, but its accuracy declined for extreme categories due to class imbalance. To reduce this problem, the SMOTE-Tomek was used to improve minority class representation, enhancing the overall robustness of the classification model.

The framework provides PM<sub>2.5</sub> concentration together with AQI categories and health-oriented remarks, supporting **policy, decisions, public health advisories, and early warning systems**. By turning results into easy-to-understand AQI labels, it **enhances public awareness of air pollution risks**. Keeping real extreme values helps the system detect extreme pollution events, improving its readiness for emergencies and quick responses.

Future work could integrate additional meteorological, geographical, and real-time data, including satellite AOD, traffic, and industrial input, to improve predictions in diverse cities. Incorporating IoT air sensors and mobile applications can support real-time forecasting and alert messages for users. Expanding the system to a national scale and combining it with deep learning models can further increase accuracy and help reduce health risks from air pollution.

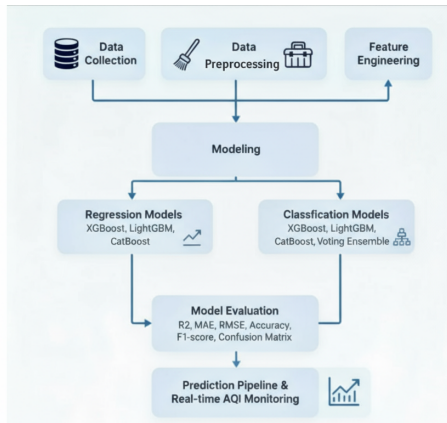


Fig. 8. Flowchart

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